

Graphing $f(x) = |x|$ and Key Attributes - Practice

Algebra II Honors | Unit 2 | Lesson 1 Practice | TEK A2.2.A

Objective: I can evaluate $f(x) = |x|$ using its piecewise definition, derive its key attributes algebraically, and reason about even functions.

Section 1 — Recall

1. Evaluate $f(-9)$ and $f(5)$ for $f(x) = |x|$ using the piecewise definition, stating which piece is used each time.

$$f(-9) = \underline{\hspace{2cm}} \text{ (uses } -x\text{)} \quad f(5) = \underline{\hspace{2cm}} \text{ (uses } x\text{)}$$

2. State the domain, range, and vertex of $f(x) = |x|$, then show algebraically that $f(-x) = f(x)$.

Domain: _____ Range: _____ Vertex: _____

$$f(-x) = |-x| = \underline{\hspace{2cm}}$$

Section 2 — Apply

3. Using the piecewise definition (not the graph), derive the range of $f(x) = |x|$.

$$x \geq 0: y \geq \underline{\hspace{2cm}} \quad x < 0: y \underline{\hspace{2cm}} \quad \text{Range: } \underline{\hspace{2cm}}$$

4. Determine algebraically whether $f(x) = |x|$ is even, odd, or neither, and justify.

$$f(-x) = |-x| = \underline{\hspace{2cm}} \rightarrow \underline{\hspace{2cm}}$$

Section 3 — Challenge

5. Let $h(x) = |x| - |-x|$. Evaluate $h(-4)$ and $h(7)$, then determine what $h(x)$ simplifies to for every real number x .

$$h(-4) = \underline{\hspace{2cm}} \quad h(7) = \underline{\hspace{2cm}} \quad h(x) = \underline{\hspace{2cm}}$$

6. A function $k(x)$ satisfies $k(-x) = k(x)$ for every x , just like $f(x) = |x|$. Does this alone guarantee $k(x) = |x|$? Give a different example of an even function to support your answer.

No _____

Section 4 — Explain It

7. Explain why the piecewise definition guarantees $f(x) = |x|$ can never produce a negative output. Reference both pieces of the definition.
